MERGE: Multimodal Emotion-Driven Rhythmic Lyric Generation Engine

Abhay Sharma **Ankit Kumar** **Hritik Ranjan** **Vikash Kumar**

Abstract

Creative language generation has gained increasing attention in natural language processing, particularly for tasks such as poetry and song lyric generation. However, most existing systems focus either on structural constraints such as rhyme and meter or on semantic control mechanisms such as sentiment, topic, or style. The integration of multimodal emotional signals with rhythm-aware lyric generation remains relatively underexplored.

In this work, we propose **MERGE** (Multimodal Emotion-Driven Rhythmic Lyric Generation Engine), a framework for generating rhythm-aware song lyrics conditioned on emotional states inferred from facial expressions. The system first detects the user's emotional state using a facial emotion recognition model and then conditions GPT-2 on the predicted emotion to guide lyric generation. To maintain lyrical structure, we employ a rhythm-aware reranking strategy that selects candidate lines based on syllable counts and rhyme patterns.

We further introduce **PHREM**, a composite evaluation metric designed for lyric generation that measures rhythmic consistency, emotional alignment, lexical diversity, and semantic coherence. MERGE provides a framework for exploring emotion-conditioned lyric generation while preserving structural properties of song lyrics.

1 Introduction

The motivation of this project is to develop an emotion-aware system capable of generating creative musical content based on a user's emotional state inferred from sensory inputs. Direct music synthesis conditioned on physiological or multimodal sensor data poses significant challenges due to limited dataset availability, the high dimensionality of audio generation, and the complex alignment between emotional intent and musical structure (Cao et al., 2022). To preserve the objective of emotion-driven creative generation while remaining within the methodological scope of natural language processing (NLP), this work reformulates the problem as emotion-conditioned lyric generation.

Facial emotion recognition provides a practical and widely studied modality for sensing emotional states (Goodfellow et al., 2013; Li et al., 2017). At the same time, song lyrics serve as a primary textual medium through which emotional expression is conveyed in music. These properties make lyrics a suitable domain for studying emotion-controlled and structurally constrained language generation. Prior research in NLP has explored emotion-conditioned text generation as well as structural constraints such as rhyme schemes and syllable patterns in poetry and lyric generation. However, these directions have largely been investigated independently.

This work leverages large-scale lyric corpora obtained from the Genius lyric dataset (Lim and Benson, 2021), which provides substantially larger collections of song lyrics. Since explicit emotion annotations are not available for this dataset, we construct emotion labels using a scalable annotation pipeline that aggregates predictions from a GoEmotions-based classifier and maps fine-grained emotional categories into the six facial emotion recognition (FER) classes through normalization and weighted line-level aggregation. This procedure enables the creation of large-scale emotion-labeled lyric datasets suitable for training emotion-conditioned language models.

While prior work has examined controlled text generation and rhythm-aware lyric generation separately, the integration of facial emotion signals as control inputs for rhythm-aware lyric generation remains relatively underexplored, particularly in English-language songwriting. Bridging this gap requires a framework that can incorporate visual emotional cues into the generation process while

maintaining the structural characteristics that define song lyrics.

In this work, we propose **MERGE (Multimodal Emotion-Driven Rhythmic Lyric Generation Engine)**, a multimodal framework that connects facial emotion recognition with rhythm-aware lyric generation. MERGE uses facial emotion recognition models to infer emotional states from visual inputs and employs these signals to condition a transformer-based language model trained on large-scale lyric corpora. To ensure structural coherence, the generation process incorporates rhythm-aware decoding constraints based on syllable counts and rhyme schemes.

This work presents MERGE, a multimodal framework for emotion-conditioned lyric generation that integrates facial emotion recognition with rhythm-aware text generation.

The contributions of this work are fourfold: (1) MERGE, a unified framework that connects facial emotion recognition with lyric generation; (2) a scalable emotion annotation pipeline that maps fine-grained GoEmotions predictions to FER categories using normalization and weighted aggregation, enabling large-scale labeling of lyric corpora; (3) a rhythm-aware reranking strategy that promotes syllabic and rhyming structure during generation; and (4) PHREM, a composite evaluation metric for assessing rhythmic consistency, emotional alignment, and overall lyric quality.

The source code and implementation details for this project are available at the following link.[1]

2 Related Work

Facial Emotion Recognition. Facial emotion recognition has achieved strong performance using deep convolutional neural networks, with architectures such as ResNet and MobileNet commonly evaluated on benchmark datasets including FER-2013 and RAF-DB (Goodfellow et al., 2013; Li et al., 2017). The FER-2013 dataset contains over 35,000 grayscale facial images annotated with basic emotion categories and remains a widely used benchmark due to its scale and accessibility (Goodfellow et al., 2013). Recent work has explored transformer-based architectures and multimodal fusion methods, improving the modeling of contextual emotional signals (Zhang et al., 2023). These advances make facial emotion recognition a reliable component for downstream generative systems.

Controlled Text Generation. Controlled text generation aims to steer language models toward desired attributes such as sentiment, topic, or style. Early approaches relied on conditional fine-tuning and control tokens (Radford et al., 2019), while more recent methods explore attribute-guided generation for aligning outputs with semantic properties such as emotional tone (Hu et al., 2023). These approaches demonstrate that external control signals can effectively guide generative language models.

Lyric and Poetry Generation. Prior work on poetry and lyric generation has focused on enforcing structural constraints such as rhyme, meter, and syllable patterns. Early approaches used rule-based methods and phonetic resources like the CMU Pronouncing Dictionary (Ghazvininejad et al., 2016; Gonçalo Oliveira, 2024). Neural models later incorporated such constraints to produce more coherent poetic text. Recent systems further integrate musical or prosodic signals to guide generation. For instance, SongRewriter (Sun et al., 2023) uses multi-level masking for controllable lyric generation, while XAI-Lyricist (Liang et al., 2024) incorporates prosody to improve singability and interpretability. However, these approaches primarily focus on structural or musical guidance and do not explicitly study the interaction between rhythmic constraints and emotion-conditioned generation.

Collaborative and Multimodal Generation Systems. Recent work has explored multi-agent and multimodal approaches for lyric generation. Liu et al. (Liu and Liu, 2024) propose a multi-agent system for handling rhyme, syllable control, and alignment tasks, while Ram et al. (Ram et al., 2021) use multitask learning with T5 to model lyrical constraints. More broadly, multimodal generative models such as MusicLM (Huang et al., 2023) demonstrate the potential of combining semantic signals with generative models, although they primarily focus on audio generation rather than lyrics.

Despite these advances, existing systems either emphasize structural generation or rely on musical signals such as melody and prosody. The integration of visual emotion signals such as facial expressions with rhythm-aware lyric generation remains underexplored, particularly in English-language songwriting. MERGE addresses this gap by providing a framework that connects facial emotion recognition with rhythm-aware lyric generation.

[1] https://github.com/IIITH-2025-27/NLP_Project

3 Dataset Details

3.1 Lyric Corpus

Lyrics are sourced from the **Genius** corpus (Lim and Benson, 2021), which contains songs spanning many languages. After filtering for English-language content, approximately **3.2 million** songs remain. A cleaned version of this corpus is publicly available on Kaggle.[2]

3.2 Facial Expression Datasets

The FER component is trained on a unified corpus comprising three publicly available datasets:

- **FER-2013** (Goodfellow et al., 2013): 35,887 grayscale 48×48 images, seven original classes (disgust excluded, yielding six classes).
- **RAF-DB** (Li et al., 2017): ∼15,339 real-world images with basic and compound emotion labels (disgust excluded), It was found on kaggle.[3]
- **AffectNet** (Mollahosseini et al., 2017): ∼450,000 images annotated with eight emotions. But out of them only ∼30,000 was used for training (disgust excluded), which were found on kaggle.[4]

3.3 Dataset Construction and Emotion Mapping

Large-scale lyric datasets typically lack explicit emotion annotations, which makes it difficult to train emotion-conditioned language models. To address this limitation, we construct an emotion-labeled lyric dataset using an automatic annotation pipeline as shown in Figure 1.

Lyrics are collected from large public lyric corpora, specifically the Genius dataset (Lim and Benson, 2021), which contains lyrics in many languages out of which approx 3.2 million belongs to English language. Emotional signals are inferred using a pretrained emotion classifier based on the RoBERTa architecture(Lowe, 2023), which is fine-tuned on the GoEmotions dataset (Demszky et al., 2020). The GoEmotions label space contains 28 categories, consisting of 27 fine-grained emotion labels and one neutral category. For each input lyric line, the classifier outputs a probability distribution over these 28 labels, which serves as the basis for constructing the emotion representations used in the subsequent stages of the pipeline.

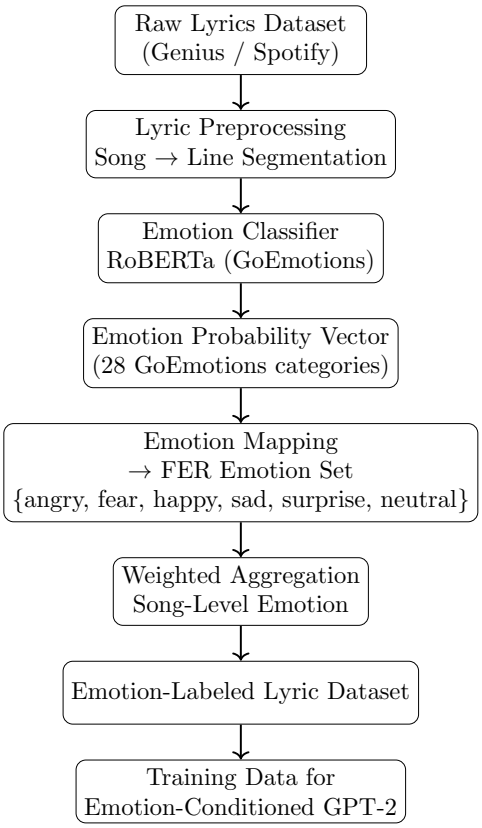

Figure 1: Pipeline for constructing the emotion-labeled lyric dataset.

To ensure compatibility between textual emotions and facial emotion recognition outputs, the fine-grained emotion space is mapped into a smaller set of basic emotions, inspired by the circumplex model of affect and theories of basic emotions (Russell, 1980; Ekman, 1992). The emotion space is reduced to the following set

$$\mathcal{E} = \{\text{angry}, \text{fear}, \text{happy}, \text{sad}, \text{surprise}, \text{neutral}\}.$$

The disgust category is excluded for two reasons. First, explicit expressions of disgust rarely occur in song lyrics and are often linguistically expressed as sadness or disappointment. Second, the FER-2013 facial emotion dataset contains very few samples labeled as disgust, resulting in severe class imbalance during model training. To maintain a consistent emotion space across modalities, disgust-related signals are therefore merged into

[2] https://www.kaggle.com/datasets/ankit477/genius-dataset-cleaned
[3] https://www.kaggle.com/datasets/shuvoalok/raf-db-dataset
[4] https://www.kaggle.com/datasets/mstjebashazida/affectnet

the sadness category. The created dataset is being hosted on kaggle(Kumar et al., 2025).

3.4 Lyric Dataset and Emotion Labeling

Each song is segmented into individual lyric lines. Let a song contain n lyric lines, where the index of a line is denoted by

$$i \in \{1, 2, \ldots, n\}.$$

For each lyric line i, the GoEmotions classifier produces a probability vector over the 28 emotion labels:

$$\mathbf{p}_i = \left[p_i^{(1)}, p_i^{(2)}, \ldots, p_i^{(28)}\right] \in \mathbb{R}^{28},$$

where $p_i^{(e)}$ denotes the probability assigned to label e and satisfies

$$0 \leq p_i^{(e)} \leq 1.$$

To obtain emotion representations compatible with facial emotion recognition categories, the fine-grained emotion labels are grouped into six emotion sets:

$$\begin{aligned}
A &= \{\text{anger}, \text{annoyance}, \text{disapproval}\},\\
F &= \{\text{fear}, \text{nervousness}\},\\
H &= \{\text{admiration}, \text{amusement}, \text{approval}, \text{caring},\\
&\quad \text{desire}, \text{excitement}, \text{gratitude}, \text{joy}, \text{optimism},\\
&\quad \text{love}, \text{pride}, \text{relief}\},\\
S &= \{\text{sadness}, \text{grief}, \text{disappointment}, \text{remorse},\\
&\quad \text{disgust}\},\\
U &= \{\text{surprise}, \text{realization}\},\\
N &= \{\text{neutral}, \text{curiosity}, \text{confusion}\}.
\end{aligned}$$

Using these groups, the emotion score for each category is computed by summing probabilities belonging to the same group. For example, the anger score for line i is defined as

$$q_i^{\text{angry}} = \sum_{e \in A} p_i^{(e)}.$$

Applying the same procedure to all emotion groups produces a six-dimensional emotion vector for lyric line i:

$$\mathbf{q}_i = \left[q_i^{\text{angry}}, q_i^{\text{fear}}, q_i^{\text{happy}}, q_i^{\text{sad}}, q_i^{\text{surprise}}, q_i^{\text{neutral}}\right].$$

Because multiple probabilities are aggregated, the resulting vector is normalized to obtain a valid probability distribution:

$$\mathbf{q}_i = \frac{\mathbf{q}_i}{\sum_{k=1}^{6} q_i^{(k)}}.$$

Lyrics often contain narrative lines with little emotional content. To reduce their influence, each line is assigned an emotional weight defined as

$$w_i = 1 - q_i^{\text{neutral}},$$

where w_i represents the emotional intensity of lyric line i.

The final emotion representation of a song is computed by aggregating the line-level emotion vectors using weighted averaging:

$$\mathbf{E}_{\text{song}} = \frac{\sum_{i=1}^{n} w_i \mathbf{q}_i}{\sum_{i=1}^{n} w_i}.$$

The dominant emotion of the song is defined as

$$e^* = \arg\max_k \mathbf{E}_{\text{song}}^{(k)},$$

where $\mathbf{E}_{\text{song}}^{(k)}$ denotes the score of emotion category k.

4 Methodology

This work proposes an end-to-end multimodal framework that connects facial emotion recognition with emotion-conditioned and rhythm-aware lyric generation.

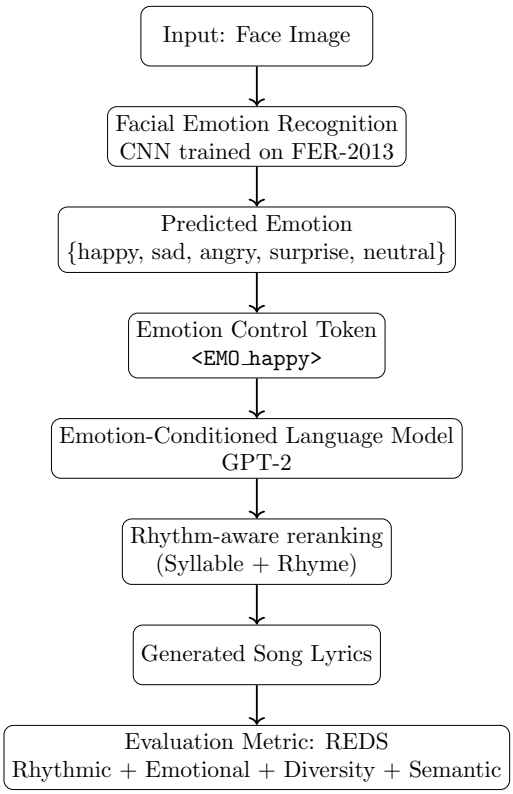

Figure 2: Overview of the MERGE framework.

The system (as shown in Figure 2) first detects emotional states from facial inputs and then uses these signals to guide lyric generation while enforcing linguistic and rhythmic constraints such as rhyme patterns and syllable structure.

4.1 Facial Emotion Conditioning

The MERGE framework uses facial expressions as the emotional input signal for lyric generation. Given a facial image, a facial emotion recognition model predicts the emotional state of the user (as shown in Figure 3), which is then used to condition the lyric generation process.

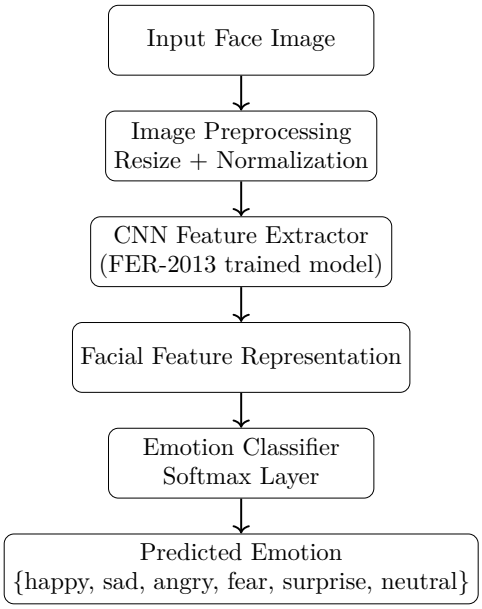

Figure 3: Pipeline of the facial emotion recognition module.

The facial emotion recognition model is trained on the datasets described in Section 3.2.

Let I denote the input facial image and let f_θ represent the facial emotion recognition model parameterized by θ. The model produces a vector of logits

$$\mathbf{z} = f_\theta(I)$$

where $\mathbf{z} \in \mathbb{R}^6$ represents the unnormalized scores corresponding to the six emotion categories.

The probability distribution over emotions is computed using the softmax function

$$P(e_k \mid I) = \frac{\exp(z_k)}{\sum_{j=1}^{6} \exp(z_j)}$$

where $e_k \in \mathcal{E}$ denotes the k-th emotion category and z_k represents the corresponding logit value.

The predicted facial emotion is obtained by selecting the class with the highest probability

$$e^* = \arg\max_{e_k \in \mathcal{E}} P(e_k \mid I)$$

The predicted emotion e^* is then used as the conditioning signal for the lyric generation model. During inference, the specified emotion is encoded as a control token (for example <sad>) and prepended to the input sequence of the language model. This allows the model to generate lyrics that are aligned with the emotional state inferred from the user's facial expression.

4.2 Emotion-Conditioned Language Model

Lyric generation is formulated as a controlled text generation task in which emotional context influences the semantic content of generated lyrics. We fine-tune the GPT-2 language model (Radford et al., 2019) on the emotion-labeled lyric dataset constructed in the previous subsection.

Emotion conditioning is implemented using explicit control tokens representing the target emotion category. During training, an emotion token (for example <sad>) is prepended to each lyric sequence so that the model can learn associations between emotional states and linguistic patterns.

Let

$$x = (x_1, x_2, \ldots, x_T)$$

denote a lyric token sequence of length T, and let e represent the conditioning emotion label. The model learns the conditional probability

$$P(x \mid e) = \prod_{t=1}^{T} P(x_t \mid x_{<t}, e),$$

where x_t denotes the token generated at step t and $x_{<t}$ represents the previously generated tokens.

The model is trained using the standard cross-entropy objective for next-token prediction.

4.3 Rhythm-Aware Reranking

To encourage rhythmic and rhyming structure during lyric generation, we employ a rhythm-aware reranking strategy. The CMU Pronouncing Dictionary provides phonetic representations that enable syllable counting and rhyme detection (CMU, 2014).

Given an input emotion, multiple candidate lyric lines are generated using stochastic sampling. Each candidate is then evaluated using a scoring function

that incorporates both syllabic meter and rhyme consistency.

Syllable counts are computed using CMU-based pronunciations when available, with a fallback heuristic for out-of-vocabulary words. A meter score penalizes deviations from a predefined target syllable count. Formally, given a line L and target syllable count T, the meter score is defined as:

$$\text{MeterScore}(L) = -\left|\sum_{w \in L} \text{syllables}(w) - T\right|.$$

Rhyme consistency is evaluated using phonetic rhyme segments derived from CMU pronunciations. A binary rhyme score is assigned based on whether the final word of a candidate line matches the rhyme pattern of a reference word:

$$\text{RhymeScore}(L) = \begin{cases} 1, & \text{if rhyme matches} \\ 0, & \text{otherwise} \end{cases}$$

The overall score is computed as a weighted combination of meter and rhyme:

$$\text{Score}(L) = \text{MeterScore}(L) + \lambda \cdot \text{RhymeScore}(L),$$

where λ controls the relative importance of rhyme (set to 5 in our implementation).

Additionally, low-quality candidates are filtered using simple heuristics that remove excessively short, repetitive, or filler-dominated lines.

The final output is selected as the highest-scoring candidate, enabling the model to produce rhythmically and phonetically structured lyrics without modifying the underlying decoding process. This formulation prioritizes rhythmic consistency while softly encouraging rhyme structure, allowing flexibility when strict constraint satisfaction is not achievable.

5 Evaluation Metric: PHREM

Existing metrics such as BLEU and ROUGE measure lexical overlap but are insufficient for evaluating song lyrics, which require assessment of phonetic flow, rhythmic consistency, emotional coherence, and semantic structure. We introduce **PHREM** (Phonetic Rhythm Emotion Metric), a composite evaluation framework designed for lyric quality assessment in the context of emotion-conditioned generation.

PHREM integrates seven complementary components, each targeting a distinct dimension of lyric quality. The base score is computed as a weighted sum:

$$\text{PHREM} = \sum_{i=1}^{7} \alpha_i c_i \tag{1}$$

where weights are: $\alpha_{\text{ppfs}} = 0.12$, $\alpha_{\text{rsfs}} = 0.13$, $\alpha_{\text{eas}} = 0.22$, $\alpha_{\text{hqcs}} = 0.13$, $\alpha_{\text{rcs}} = 0.10$, $\alpha_{\text{mcs}} = 0.20$, $\alpha_{\text{dps}} = 0.10$. All components are normalized to $[0, 1]$, ensuring $\text{PHREM}_{\text{final}} \in [0, 1]$. Component weights were determined through human judgment, where annotators assessed the relative contribution of each dimension to overall lyric quality.

PPFS: Phonetic Pattern Flow Score. PPFS evaluates phoneme-level smoothness by penalizing abrupt vowel-consonant transitions and stress mismatches between adjacent phonemes. Phoneme sequences are extracted using the CMU Pronouncing Dictionary, with G2P as a fallback for out-of-vocabulary words. For adjacent phonemes p_i and p_{i+1}:

$$\begin{aligned} c(p_i, p_{i+1}) = 0.6 \cdot \mathbf{1}[v(p_i) \neq v(p_{i+1})] \\ + 0.4 \cdot |s_i - s_{i+1}| \end{aligned} \tag{2}$$

where $v(p)$ indicates whether phoneme p is a vowel, $\mathbf{1}[\cdot]$ indicates a vowel-consonant transition, and $|s_i - s_{i+1}|$ captures stress mismatch. The final score is $\text{PPFS} = \exp(-\bar{c})$, where $\bar{c}$ is the mean transition cost across all adjacent phoneme pairs.

RSFS: Rhythmic Structure Flow Score. RSFS measures rhythmic consistency across lines by evaluating syllable count uniformity and stress pattern regularity. Syllable consistency is measured relative to the median line length:

$$R_s = 1 - \frac{1}{n} \sum_{i=1}^{n} \frac{|s_i - \text{med}(s)|}{\text{med}(s)} \tag{3}$$

Stress consistency R_{stress} is computed by comparing stress sequences of consecutive lines over their overlapping syllable positions. The final score is $\text{RSFS} = 0.7R_s + 0.3R_{\text{stress}}$, weighting syllable uniformity more heavily.

EAS: Emotion Arc Score. EAS measures emotional consistency across the lyric sequence. For each line i, a RoBERTa-based GoEmotions classifier produces a probability vector that is mapped and normalized to six FER emotion categories,

yielding $\mathbf{q}_i \in [0,1]^6$. Transitions between consecutive lines are measured using Jensen-Shannon divergence:

$$d_i = \frac{1}{2}\left[\mathrm{KL}(\mathbf{q}_i \| \mathbf{m}) + \mathrm{KL}(\mathbf{q}_{i+1} \| \mathbf{m})\right] \quad (4)$$

where $\mathbf{m} = \frac{1}{2}(\mathbf{q}_i + \mathbf{q}_{i+1})$. Each transition is weighted by emotional intensity $w_i = 1 - q_i^{\text{neutral}}$ to reduce the influence of narratively neutral lines. The final score is $\text{EAS} = \exp(-\bar{d}_w)$, where $\bar{d}_w$ is the weighted mean divergence. Lower divergence between consecutive lines yields higher scores, rewarding smooth emotional progression. EAS receives the highest weight ($\alpha_{\text{eas}} = 0.22$) as emotional coherence is central to the generation objective of MERGE.

HQCS: Hook Quality and Catchiness Score. HQCS rewards the presence of repeated n-gram patterns that contribute to memorability. All 3-grams are extracted and those appearing more than once are treated as candidate hooks. The score combines salience (maximum hook frequency relative to total n-grams), coverage (total hook occurrences relative to total n-grams), and flow quality (mean of PPFS and RSFS):

$$\text{HQCS} = 0.4 \cdot \text{sal} + 0.3 \cdot \text{cov} + 0.3 \cdot \text{flow} \quad (5)$$

A diversity penalty $\exp(-k \cdot (1 - \text{diversity}))$ is incorporated to prevent degenerate repetition from being rewarded. When no repeated n-grams are present, HQCS falls back to the mean of PPFS and RSFS.

RCS: Rhyme Consistency Score. RCS evaluates end-rhyme quality by comparing the phonetic endings of consecutive line-final words. The rhyme region is defined from the last stressed vowel onward, extracted using the CMU Pronouncing Dictionary. Similarity between consecutive rhyme sequences is computed by matching phonemes backward from the word ending. RCS is the mean similarity across all consecutive line pairs. We note that this approach captures direct phonetic rhyme but does not model rhyme scheme structure (e.g., AABB, ABAB) or slant rhymes.

MCS: Motif Consistency Score. MCS measures thematic coherence across lines using sentence embeddings from `all-MiniLM-L6-v2`. Both local coherence (between adjacent lines) and global coherence (across sampled non-adjacent pairs) are considered:

$$\text{MCS} = 0.7 \cdot L + 0.3 \cdot G \quad (6)$$

where L is the mean cosine similarity of adjacent line embeddings and G is the mean cosine similarity of a random sample of non-adjacent pairs. This formulation rewards lyrics that maintain a consistent topic or theme throughout, complementing EAS which focuses on emotional rather than semantic continuity.

DPS: Degeneracy Penalty Score. DPS penalizes excessive lexical repetition, which is a common failure mode in autoregressive language model outputs. The penalty is defined as:

$$\text{DPS} = \exp(-3.0 \cdot r) \quad (7)$$

where $r = \frac{\text{max word frequency}}{\text{total words}}$. This decays sharply when a single word dominates the output, while allowing natural repetition in well-formed lyrics.

LPS: Length Penalty Score. LPS acts as a multiplicative gate to penalize outputs that are too short to constitute meaningful lyrics. Very short outputs may achieve artificially high scores on individual components due to limited evidence. The penalty is:

$$\text{LPS} = \begin{cases} 1.0 & \text{if } n \geq 8 \\ \left(\frac{n}{8}\right)^2 & \text{otherwise} \end{cases} \quad (8)$$

where n is the number of lyric lines. The final PHREM score is obtained by multiplying the weighted sum by LPS:

$$\text{PHREM}_{\text{final}} = \text{PHREM} \times \text{LPS} \quad (9)$$

This multiplicative formulation ensures that structural deficiencies (insufficient length) override component-level scores rather than being diluted by averaging. The threshold of $n = 8$ lines and steepness of 2.0 are chosen as reasonable priors for a minimal song verse; these hyperparameters were not empirically tuned. All components operate on $[0,1]$ scales, making PHREM directly interpretable as a normalized lyric quality score.

6 Experiments and Training Setup

6.1 Facial Emotion Recognition

To improve generalization, DenseNet-121 is trained on a unified corpus comprising FER-2013 (35,887 images), RAF-DB (∼15,000 images), and AffectNet (∼30,000 images) mapped into a six-class label space with 80/10/10 split and MTCNN face filtering (Zhang et al., 2016). The training set is expanded using domain-bridging augmentations (RandomGrayscale $p = 0.3$, GaussianBlur, ColorJitter, RandomAffine, RandomPerspective) for improved generalization across datasets. DenseNet-121 is trained on Kaggle GPU resources with progressive unfreezing over 30 epochs (5 warmup, Cosine Annealing scheduler, head first then backbone lr $= 10^{-4}$, head lr $= 10^{-3}$), Focal Loss (Lin et al., 2017) ($\gamma = 2$) with label smoothing ($\varepsilon = 0.1$). Final predictions employ test-time augmentation to generate multiple prediction sets, which are then aggregated via averaging to improve robustness and generalization across the unified dataset.

6.2 Language Model Fine-Tuning

A GPT-2 language model was fine-tuned on an emotion-labeled lyric dataset using a control-token-based conditioning strategy. Each training example was prepended with a special emotion token (e.g., `<sad>`, `<happy>`) to enable controllable generation within a single unified model.

The model was trained using the AdamW optimizer with a learning rate of 5×10^{-5} for 3 epochs. Training was performed with a batch size of 32 and a maximum sequence length of 120 tokens. To ensure balanced learning across emotions, the dataset was sampled to include up to 60,000 examples per emotion class.

During inference, lyric lines are generated using nucleus (top-p) sampling with $p = 0.9$ and a temperature of 0.85. Additional decoding constraints include a repetition penalty of 1.25 and a no-repeat n-gram size of 3 to reduce redundancy.

To further encourage rhythmic and rhyming structure, a rhythm-aware reranking mechanism is applied over multiple sampled candidates. Specifically, 5 candidate lines are generated per step and scored based on syllabic meter and rhyme consistency. The target meter is set to 8 syllables per line, and rhyme alignment is weighted using a coefficient $\lambda = 5$. The candidate with the highest combined score is selected as the final output.

All hyperparameters used for training, generation, and reranking are summarized in Table 1.

Category	**Configuration**
Training	
Model	GPT-2
Optimizer	AdamW
Learning Rate	5×10^{-5}
Epochs	3
Batch Size	32
Max Sequence Length	120
Samples per Emotion	60,000
Generation	
Decoding	Top-p (nucleus) sampling
Top-p	0.9
Temperature	0.85
Repetition Penalty	1.25
No-repeat n-gram	3
Max Tokens	30
Reranking	
Candidates	5
Target Meter	8
Rhyme Weight (λ)	5

Table 1: Hyperparameter configuration for training, generation, and rhythm-aware reranking.

7 Results

7.1 Facial Emotion Recognition

Table 2 summarises accuracy and train–test gap across different experimental phases.

Backbone	**Test Acc.**	**Train–Test Gap**
ResNet18	0.620	8–12%
ResNet34	0.626	8.5%
ResNet50	**0.674**	<2%
EfficientNet-B3	0.884*(val.Acc.)*	>10%
DenseNet-121	0.892*(val.Acc.)*	2%
Ensemble+TTA	0.751	—

Table 2: Accuracy progression from Resnet18 to DenseNet-121 with unified dataset.

On the unified validation set, DenseNet-121 achieved a validation accuracy of 0.892 and a weighted F1 score of $F1_w = 0.8920$, with a minimal train–test gap of 2%. This strong generalization across the combined FER-2013, RAF-DB, and AffectNet datasets demonstrates that the model effectively learns shared emotional patterns across diverse image sources and quality levels.

7.2 Baseline Comparison and Dataset Validation

To evaluate the quality of the constructed emotion-labeled lyric dataset, we perform a comparative analysis using the PHREM metric across multiple GPT-2 baselines.

We consider three settings: (1) a pretrained GPT-2 model without fine-tuning, (2) GPT-2 fine-tuned on the Spotify lyric dataset(Devdope, 2023), and (3) GPT-2 fine-tuned on the proposed emotion-labeled dataset. For each configuration, 100 lyric samples are generated across emotion categories, and the average PHREM score is computed.

Model	Avg. PHREM
GPT-2 (pretrained)	0.03
GPT-2 + Spotify fine-tuning	0.146
GPT-2 + Proposed dataset	**0.55**

Table 3: Baseline comparison using PHREM over 100 generated samples.

The results shown in Table 3 demonstrate a substantial improvement when fine-tuning on the proposed dataset, indicating that the constructed emotion-labeled corpus provides stronger supervision for emotion-conditioned lyric generation. Fine-tuning on the proposed dataset yields more than a 3.7$\times$ improvement over Spotify-based fine-tuning and over 18$\times$ improvement compared to the pretrained baseline.

These findings validate the effectiveness of the proposed dataset construction pipeline and highlight the importance of domain-specific, emotion-aligned data for improving lyric generation quality.

8 Analysis

Effectiveness of Emotion Conditioning. Emotion conditioning through control tokens produces clearly distinguishable lyric outputs across all six emotion categories. Qualitative inspection shows consistent thematic variation, with `<happy>` outputs favoring positive and celebratory language, while `<sad>` outputs reflect themes of loss and introspection. This indicates that control-token-based conditioning effectively guides semantic generation.

Impact of Dataset Quality. The baseline comparison using PHREM demonstrates that fine-tuning on the proposed dataset significantly improves generation quality, achieving an average score of 0.55 compared to 0.146 for Spotify-based fine-tuning and 0.03 for the pretrained model. This substantial improvement highlights the importance of large-scale, emotion-aligned supervision. The proposed annotation pipeline enables stronger alignment between textual content and emotional labels, leading to improved emotional coherence and structural quality in generated lyrics.

Role of Rhythm-Aware Reranking. The rhythm-aware reranking mechanism improves structural consistency by favoring candidates that better match target syllable counts and rhyme patterns. While constraints are not enforced during token-level decoding, reranking provides a flexible post-generation mechanism that enhances rhythmic quality without sacrificing fluency.

Limitations of Emotion Mapping. The GoEmotions-to-FER mapping introduces information loss due to the aggregation of fine-grained emotional categories into a small set of basic emotions. As a result, semantically distinct emotions may be conflated during labeling. For example, expressions of affection or dominant love may be mapped to the angry class due to overlap with intensity-related categories, while narrative or descriptive lines with low emotional intensity may be incorrectly assigned to the sad or neutral classes. This many-to-one mapping can introduce label noise, particularly in cases where emotional nuance is important, such as distinguishing between reflective sadness and neutral narration.

Ablation Insight: Emotion Mapping. Reassigning the disgust category to the angry group improves classification balance and reduces misalignment between textual and facial emotion representations, supporting the design choice of operating within a six-class emotion space.

9 Conclusion

This work introduced **MERGE**, a multimodal framework for generating emotion-conditioned and rhythm-aware song lyrics by integrating facial emotion recognition with controlled language generation. The proposed approach combines a scalable GoEmotions-to-FER annotation pipeline, a high-performing FER ensemble model, GPT-2 fine-tuning with emotion control tokens, and a rhythm-aware reranking mechanism guided by syllable and rhyme constraints.

Experimental results demonstrate that emotion-conditioned generation, when paired with a large-scale, emotion-aligned dataset, significantly improves lyric quality, achieving substantial gains in PHREM scores over both pretrained and Spotify-based baselines. The reranking strategy further enhances structural consistency without compromising linguistic fluency.

Overall, MERGE provides an effective framework for combining multimodal emotional signals

with structured lyric generation, while highlighting challenges related to coarse-grained emotion mapping and the absence of strict constraint enforcement during decoding.

10 Future Work

Several directions can further improve the MERGE framework. First, the current GoEmotions-to-FER mapping could be extended to preserve finer-grained emotional distinctions, reducing information loss and improving alignment between textual and visual emotion representations. Second, the rhythm-aware reranking mechanism can be strengthened by incorporating stricter constraint enforcement during generation, such as token-level control or beam-based structured decoding.

Additionally, the Rhyme Consistency Score (RCS) can be enhanced to model more complex rhyme structures. While the current formulation captures direct phonetic similarity between consecutive line endings, it does not account for global rhyme schemes (e.g., AABB, ABAB) or slant rhymes. Extending RCS to incorporate rhyme scheme modeling and approximate phonetic similarity would provide a more comprehensive evaluation of lyrical structure.

Finally, scaling the language model and incorporating richer multimodal inputs, such as audio or prosodic features, could further improve the quality and expressiveness of generated lyrics.

References

Qian Cao, Xu Chen, Ruihua Song, Hao Jiang, Guang Yang, and Zhao Cao. 2022. Multi-modal experience inspired ai creation. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 1445–1454.

CMU. 2014. The CMU Pronouncing Dictionary. http://www.speech.cs.cmu.edu/cgi-bin/cmudict.

Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. 2020. Goemotions: A dataset of fine-grained emotions. In *Proceedings of the 58th annual meeting of the association for computational linguistics*, pages 4040–4054.

Devdope. 2023. 900k spotify songs with lyrics, emotions & more. Kaggle.

Paul Ekman. 1992. An argument for basic emotions. *Cognition & emotion*, 6(3-4):169–200.

Marjan Ghazvininejad, Xing Shi, Yejin Choi, and Kevin Knight. 2016. Generating topical poetry. In *Proceedings of the 2016 conference on empirical methods in natural language processing*, pages 1183–1191.

Hugo Gonçalo Oliveira. 2024. Automatic generation of creative text in portuguese: an overview. *Language Resources and Evaluation*, 58(1):7–41.

Ian J Goodfellow, Dumitru Erhan, Pierre Luc Carrier, Aaron Courville, Mehdi Mirza, Ben Hamner, Will Cukierski, Yichuan Tang, David Thaler, Dong-Hyun Lee, and 1 others. 2013. Challenges in representation learning: A report on three machine learning contests. In *International conference on neural information processing*, pages 117–124. Springer.

Zhiting Hu, Yelong Shen, and 1 others. 2023. Attribute-guided text generation with large language models. *arXiv preprint arXiv:2306.10891*.

Cheng-Zhi Anna Huang, Andrea Agostinelli, Jordi Pons, and 1 others. 2023. Musiclm: Generating music from text. *arXiv preprint arXiv:2301.11325*.

Ankit Kumar, Vikash Kumar, Hritik Ranjan, and Abhay Sharma. 2025. Lyrics with mapped emotions. Accessed on March 8, 2026.

Shan Li, Weihong Deng, and JunPing Du. 2017. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2852–2861.

Qihao Liang, Xichu Ma, Finale Doshi-Velez, Brian Lim, and Ye Wang. 2024. Xai-lyricist: Improving the singability of ai-generated lyrics with prosody explanations. In *IJCAI*, pages 7877–7885.

Derek Lim and Austin R Benson. 2021. Expertise and Dynamics within Crowdsourced Musical Knowledge Curation: A Case Study of the Genius Platform. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 15, pages 373–384.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988.

Hong-Hsiang Liu and Yi-Wen Liu. 2024. Agent-driven large language models for mandarin lyric generation. In *2024 27th Conference of the Oriental COCOSDA International Committee for the Co-ordination and Standardisation of Speech Databases and Assessment Techniques (O-COCOSDA)*, pages 1–6. IEEE.

Sam Lowe. 2023. roberta-base-go_emotions.

Ali Mollahosseini, David Chan, and Mohammad H. Sarageh. 2017. AffectNet: A database for facial expression, face action unit, and face display recognition. *IEEE Transactions on Affective Computing*, 10(1):18–31.

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, and 1 others. 2019. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9.

Naveen Ram, Tanay Gummadi, Rahul Bhethanabotla, Richard J Savery, and Gil Weinberg. 2021. Say what? collaborative pop lyric generation using multitask transfer learning. In *Proceedings of the 9th international conference on human-agent interaction*, pages 165–173.

James A Russell. 1980. A circumplex model of affect. *Journal of personality and social psychology*, 39(6):1161.

Yusen Sun, Liangyou Li, Qun Liu, and Dit-Yan Yeung. 2023. Songrewriter: A chinese song rewriting system with controllable content and rhyme scheme. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 12863–12880.

Kaipeng Zhang, Zhanpeng Zhang, Zhifeng Li, and Yu Qiao. 2016. Joint face detection and alignment using multitask cascaded convolutional networks. In *IEEE Signal Processing Letters*, volume 23, pages 1499–1503. IEEE.

Yifan Zhang, Ming Chen, and 1 others. 2023. Multimodal emotion recognition with transformer-based fusion networks. *IEEE Transactions on Affective Computing*.